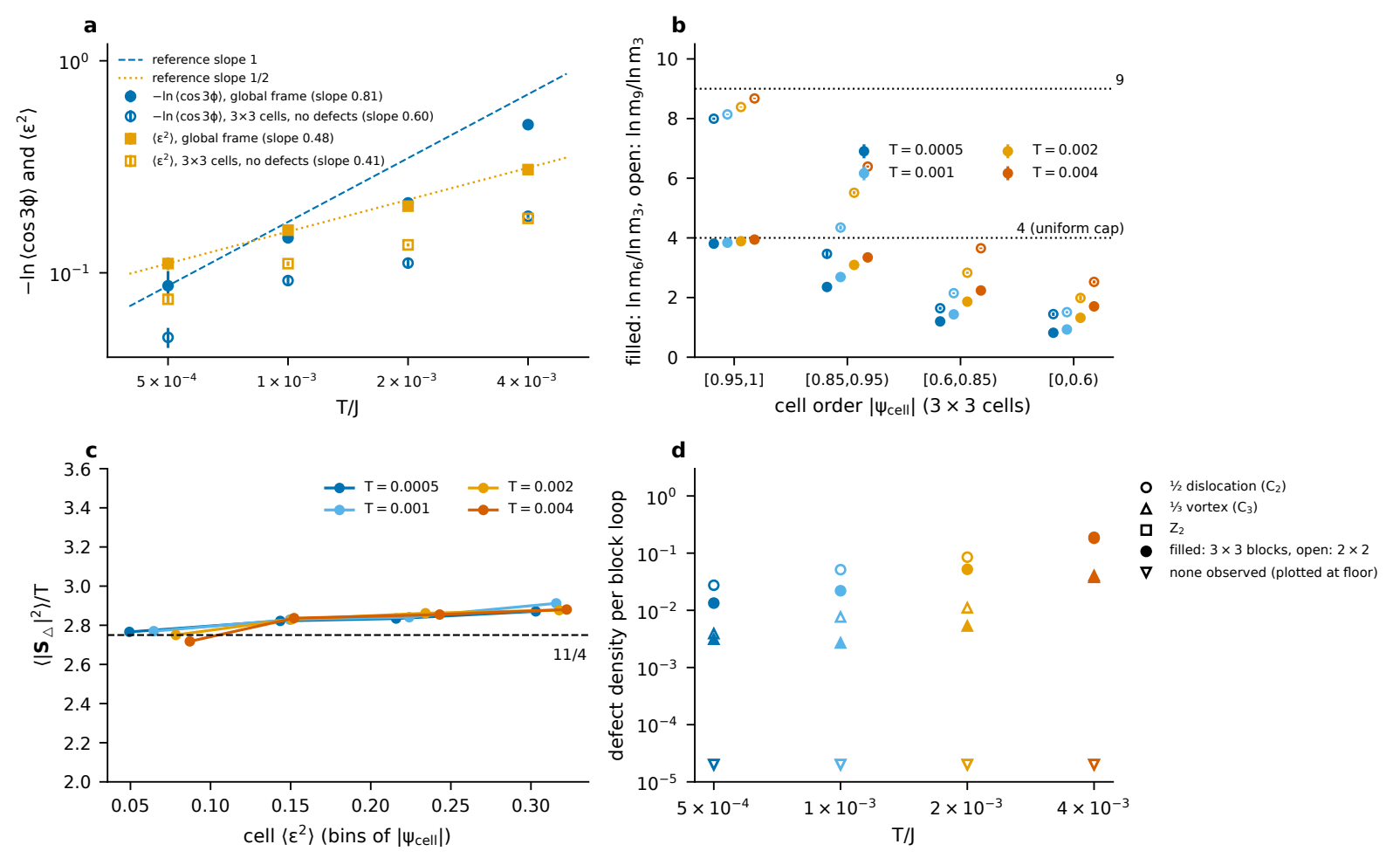
a
reference slope 1
reference slope 1/2
−ln (cos 3ϕ), global frame (slope 0.81)
−ln (cos 3ϕ), 3×3 cells, no defects (slope 0.60)
⟨ε²⟩, global frame (slope 0.48)
⟨ε²⟩, 3×3 cells, no defects (slope 0.41)
−ln (cos 3ϕ) and ⟨ε²⟩
T/J
b
filled: ln m6/ln m3, open: ln m9/ln m3
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
9
4 (uniform cap)
[0.95,1]
[0.85,0.95)
[0.6,0.85)
[0,0.6)
cell order |ψcell| (3 × 3 cells)
c
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
11/4
⟨|S△|²⟩/T
cell ⟨ε²⟩ (bins of |ψcell|)
d
defect density per block loop
T/J
½ dislocation (C2)
⅓ vortex (C3)
Z2
filled: 3 × 3 blocks, open: 2 × 2
none observed (plotted at floor)